NullWire Phase 0

A Blockchain-Coordinated Mixnet Messaging Prototype

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Abstract

We present NullWire Phase 0, a prototype architecture for blockchain-coordinated mixnet messaging. The system keeps message payloads off-chain and uses Solana only as a decentralized control plane for node registry, stake state, and reward bookkeeping. The prototype currently implements a three-hop relay path, gateway mailbox delivery for offline recipients, configurable delay profiles, and a layered sealed packet construction that removes cleartext route metadata from the wire format. We do not claim that the current prototype establishes production-grade anonymity, solves global traffic analysis, or validates a final cryptographic packet format such as Sphinx. The contribution of this work is narrower: it defines a technically consistent baseline that can be built, measured, and reviewed before stronger anonymity claims are made.

1 Introduction

Anonymous communication projects usually fail in one of two ways. Either they are architecturally weak from the start, or they make stronger claims than their implementation can justify. NullWire Phase 0 exists to avoid the second failure mode.

The prototype begins from one fixed architectural decision:

- message payloads remain off-chain;
- Solana is used as control plane only.

This cuts away the dead-end idea of putting encrypted payloads into program accounts. That approach is worse on throughput, cost, operational complexity, and visibility. A control-plane-only design is narrower but credible.

2 System Model

Phase 0 contains four components:

Component	Responsibility
Control-plane program	Registry state, stake state, reward bookkeeping
Relay nodes	Decrypt one hop layer, apply delay profile, forward inner packet
Gateway nodes	Accept submissions, receive final delivery, store mailbox entries
CLI	Register nodes, fetch snapshots, build layered packets, poll mailboxes

Each node publishes a registry record containing:

- node identifier;
- role (`RelayLayer1`, `RelayLayer2`, `RelayLayer3`, or `Gateway`);
- network endpoint;
- transport public key;
- stake state;
- last-seen timestamp; and
- reputation placeholder.

3 Packet Construction

The Phase 0 wire format no longer exposes the route in plaintext. Instead, the sender fetches transport public keys from the registry and wraps a fixed-size payload in nested encrypted directives.

Each directive contains:

- the next hop endpoint, except at the final hop;
- the mailbox name, only at the final gateway; and
- the next inner packet.

The current hop decrypts its layer and learns only the information it needs to continue forwarding. This is a real hop-by-hop sealed packet construction. It is not yet Sphinx, and Phase 0 does not claim otherwise.

4 Delay Profiles

The prototype exposes three delay profiles:

- `NoDelay`
- `ReducedPoisson`
- `LoopixLike`

These are test settings used to measure operational tradeoffs. They are not validated anonymity modes. The purpose is to gather latency and bandwidth numbers that determine whether the project should continue.

5 What the Prototype Proves

Phase 0 is intended to prove exactly four things:

1. Off-chain encrypted payload routing through a three-hop path works.
2. Solana can coordinate nodes and rewards as a control plane.
3. Latency and bandwidth overhead can be measured instead of guessed.
4. The project can present a technically consistent public story.

6 What the Prototype Does Not Prove

Phase 0 does not prove:

- resistance to global passive adversaries;
- meaningful anonymity set size;
- production-grade censorship resistance;
- anonymous admission or zero-knowledge credentials;
- Double Ratchet integration;
- mobile, browser, group, file-transfer, or voice features.

Those ideas may matter later. They are not evidence today.

7 Test Plan

The minimum evaluation loop is:

1. register three relays and two gateways;
2. fetch a directory snapshot;
3. send Alice to Bob through the sender gateway;
4. peel one layer at each relay;
5. peel the final layer at the recipient gateway;
6. store the fixed-size payload in Bob's mailbox;
7. poll the mailbox and verify delivery;
8. disable one relay and confirm failure is surfaced;
9. measure latency and bandwidth under all delay profiles.

8 Current Claim

The only clean claim supported by this repository is:

> NullWire is a prototype of a blockchain-coordinated mixnet messaging system.

Anything stronger than that belongs in future work, not in the current abstract.